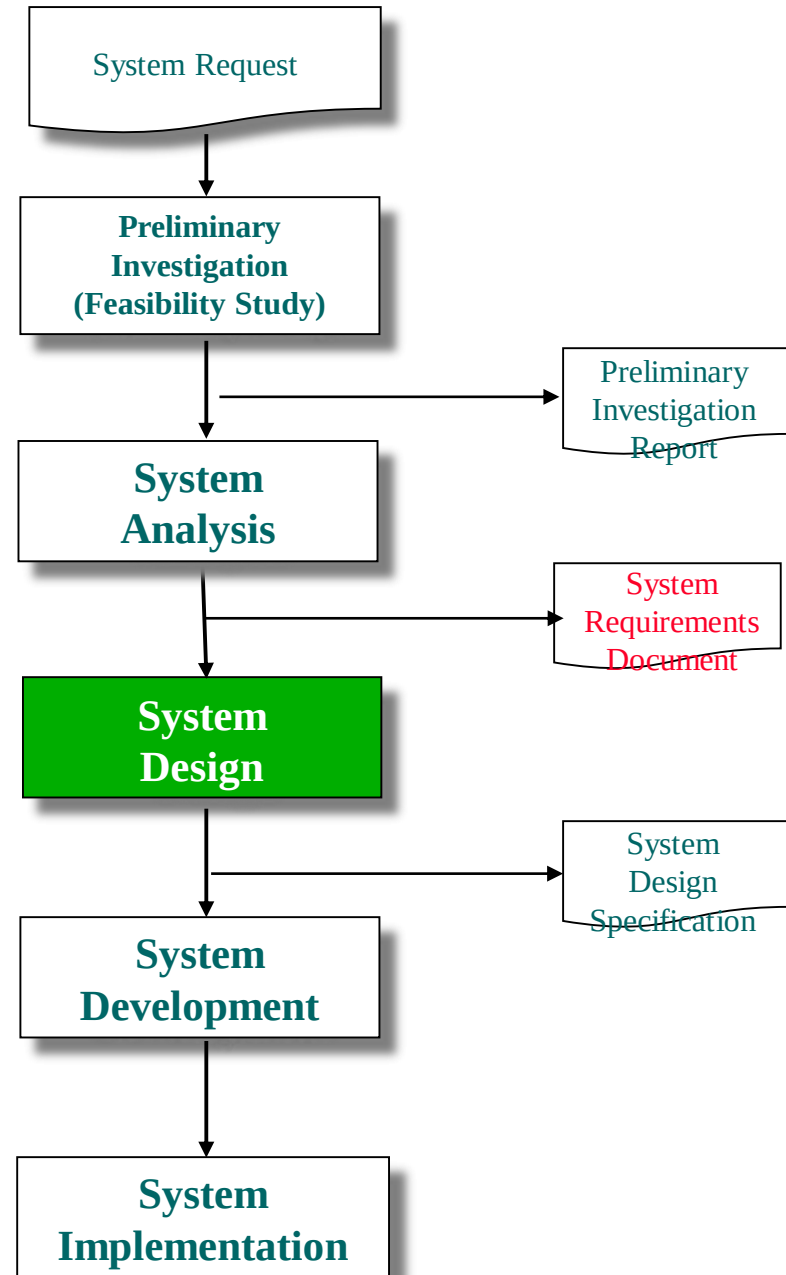


Chapter 7

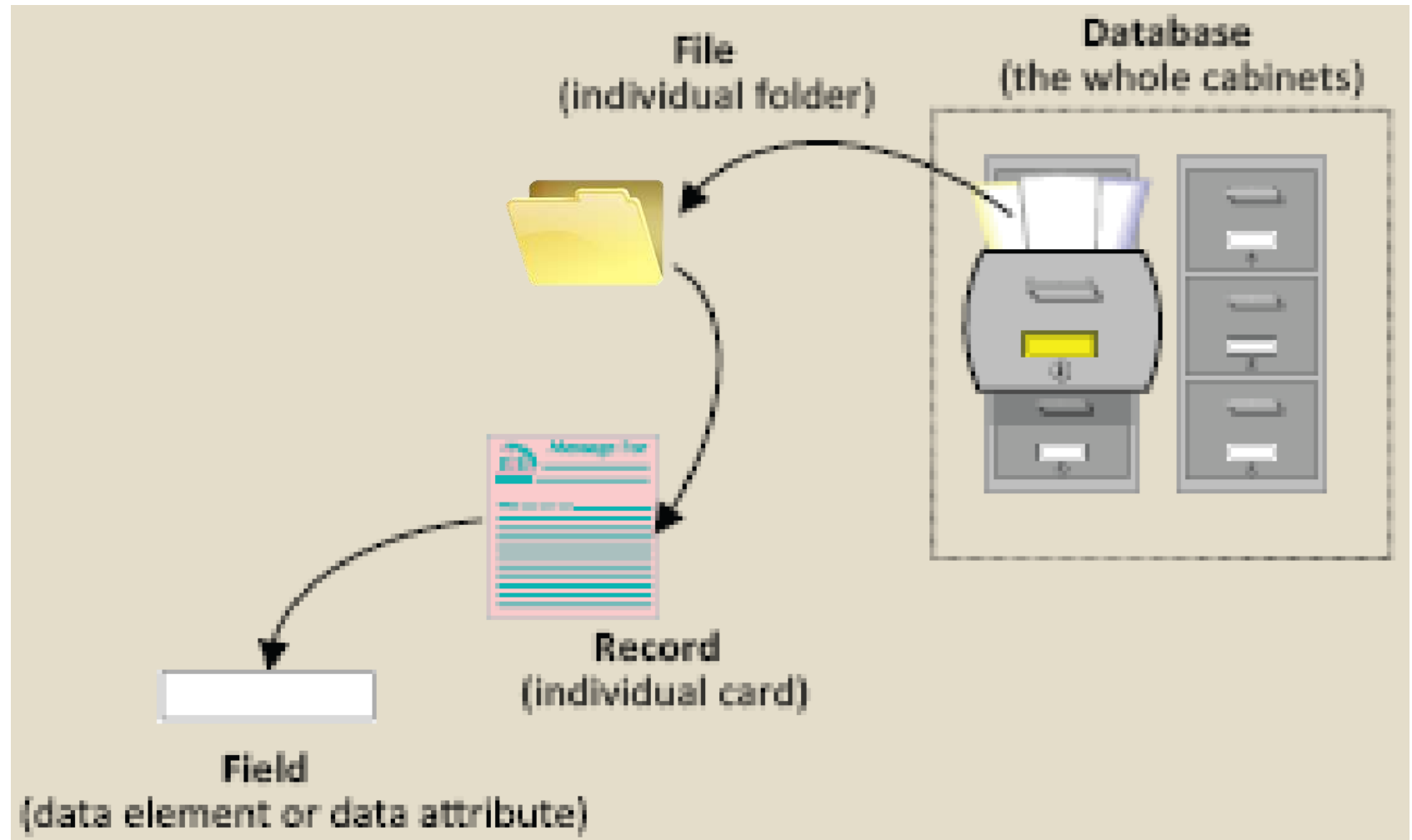
Database Design

SDLC



Database Design

Database Concept (Manual)



Database Concept (Computer)

- **Database** – consists of a number of files/tables.
- **Table (File)** – collection of related data records.
- **Record** – collection of related items of data fields.
- **Field** – contains one logical unit of data characters.

The diagram illustrates a database table structure. A table with 5 columns and 5 rows is shown. The first column, 'Employee-Number', is highlighted with a red border and labeled as the 'Key Field'. The other four columns are grouped under the label 'Non-Key Fields'. The entire table is labeled 'Table' with an arrow pointing to it. The columns are labeled 'Fields/Attributes' with an arrow pointing to them. The rows are labeled 'Records' with a bracket. The data in the table is as follows:

Employee-Number	First Name	Last Name	Telephone	Dept Number
0204	Yow Teng	Tan	74484488	435
1494	Beng Yam	Lim	73394737	634
0276	Mustapha	Ali	63748490	252
0839	James	Dykes	32227842	183

Key Field

- **Meaning** – a field used to identify and differentiate one **record** from other records in a table.

Mater Files	Key Fields
Employee	Employee No./ ID
Customer	Customer No. / ID
Stock	Stock Code

Transaction Files	Key Fields
Invoice	Invoice No.
Purchase Order	PO No.
Payment	Payment No.

Recommended:

Human – Use No. / ID

Transaction – Use No.

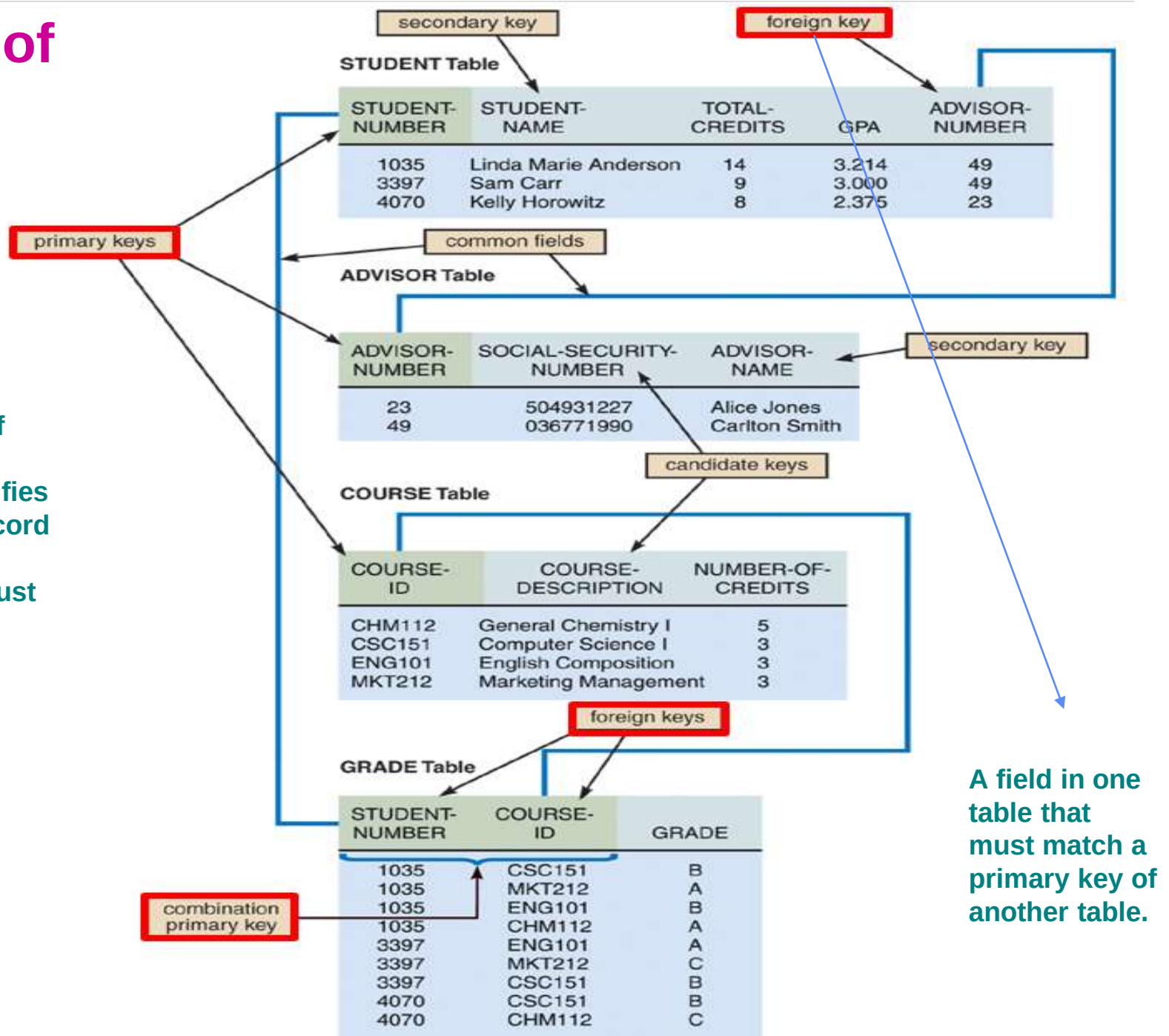
Item – Use No. / Code

Types of File/Table

- **Transaction File** – contains transaction records captured from business transactions which changes frequently with the latest transaction data eg. **Purchase Order file** contains details of a PO such as PO number, PO date. Examples: PO, SO, Invoice, Payment, RFQ, Quotation etc.
- **Master File** - contains relatively more permanent records in a table eg. a **customer file** will contain details of a customer such as customer ID, Name and Contact Address. Examples: Stock, Supplier, Student, Customer, Branches etc.
- **Organisation File** – contains organization data used to help set up the organization structure eg **Company file** contains details of a Company such as Company No, Name, Address.

Types of Keys

a field or a combination of fields that uniquely identifies a particular record of an entity. A primary key must be unique



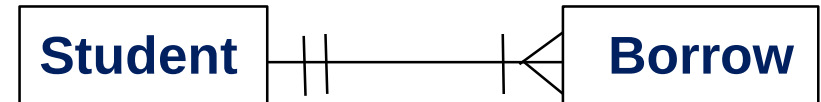
A field in one table that must match a primary key of another table.

1. Primary key

- **Definition** - a field or a combination of fields that uniquely identifies a particular record of an entity. A primary key must be unique.

- **Examples**

- Student Table – **Student ID**
- Borrow Table – **Borrow No.**



Student Table

Student ID	Name	Phone		
WM23D01	Tan Ah Meng	0335454884		
WM22B12	Lilian Wong	0445982378		

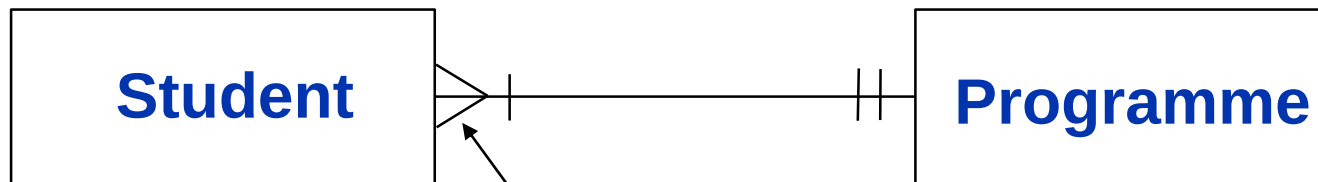
Borrow Table

Borrow No.	Date	Student ID		
000000017	24-12-23	WM22B12		
000000018	24-12-23	WM23D01		

Try fill up
other data
fields

2. Foreign Key

- **Definition** – A field in one table that must match a primary key of another table.
- **Purpose** – To establish the relationship between two tables (entities)



The '**Many**' symbol indicates that the **Foreign key** 'Programme Code' exists in the **STUDENT** entity

STUDENT (Student Number, Name, Address, Email, Telephone, Programme Code*)

PROGRAMME (Programme Code, Programme Name, Type)

Foreign Key

Student



Programme

Student Table

Student-Number	Name	Address	State	Telephone	<u>Programme Code</u>
1001	Tan Beng Hong	8, Jalan Bertuah 3, Taman Segar	SGR	78994332	BT0005
2214	Lim Chee Keong	14, Jalan 3/4G, Taman Desa	KUL	83463639	AC0002
1034	Ahmad Bin Ali	2A, Lorong 14/2, Wangsa Maju	KUL	32263840	BT0005
7435	John Dykes	13, Jalan Merlinia 3, Taman Rinting	PRK	63673764	SC0001

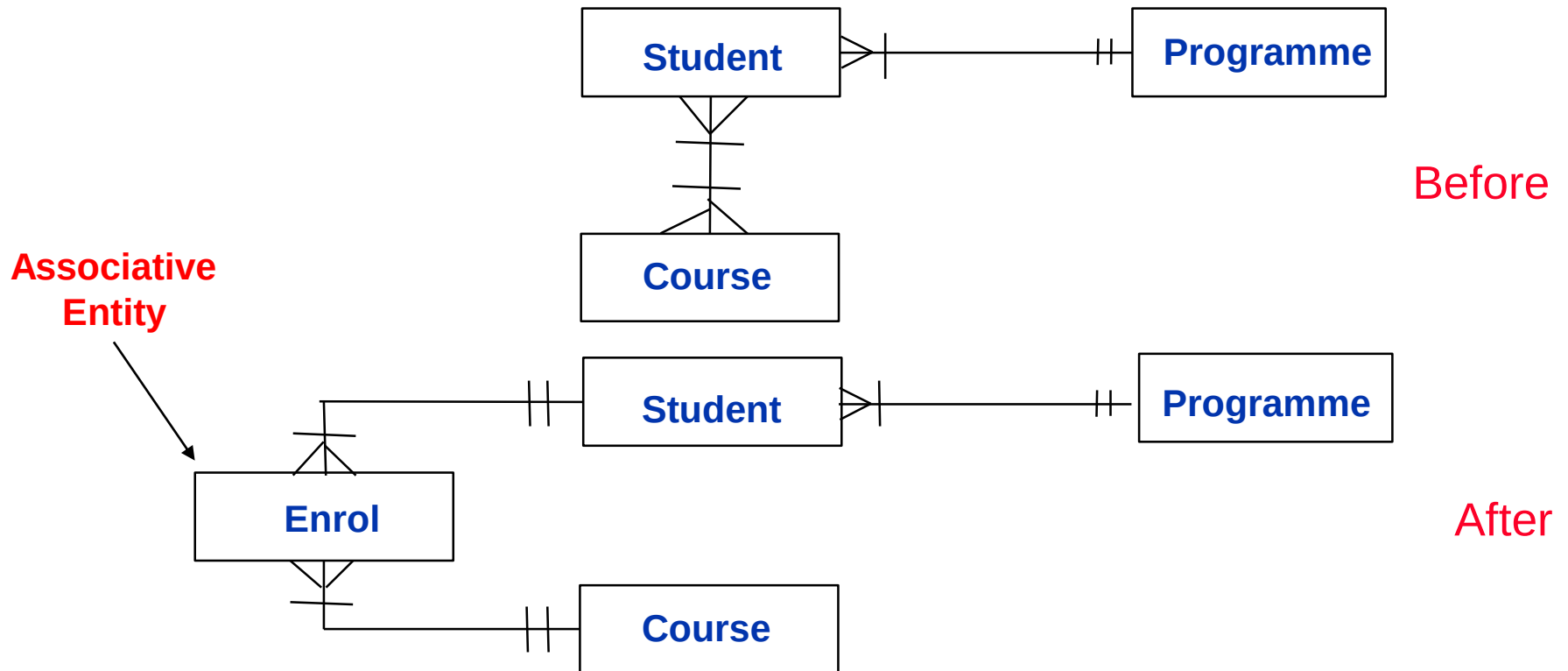
<u>Programme Code</u>	<u>Programme Name</u>	Faculty
SC0001	Degree in Biology	Sciences
BT0005	Degree in Engineering	Building Technology
AC0002	Degree in Accounting	Accountancy
IT0004	Degree in Compute Science	Information Technology
SC0002	Degree in Pure Physics	Sciences
MD0003	Degree in Medicine	Medicine

Foreign Key

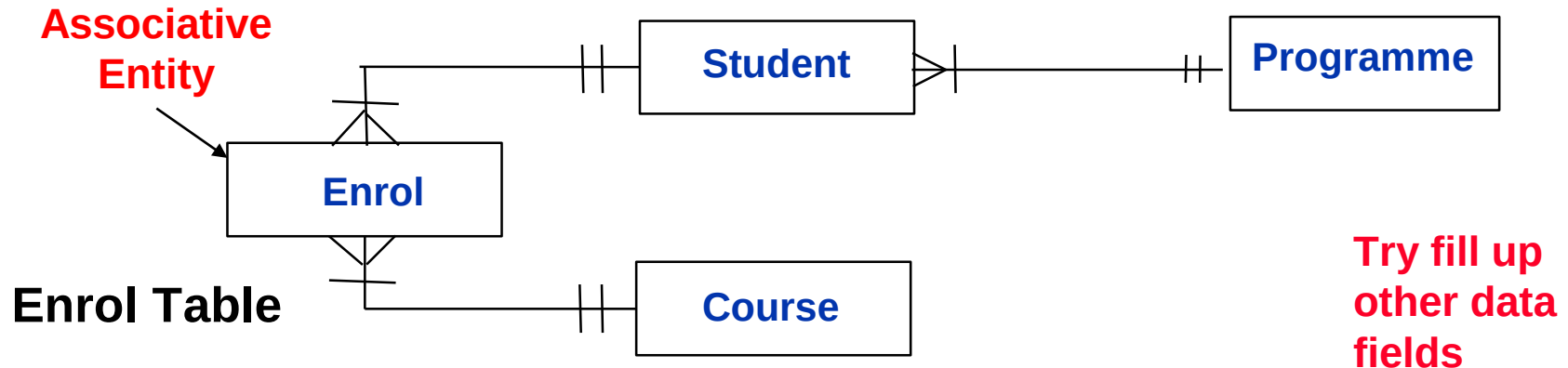
Programme
Table

3. Compound Key

- **Definition** – created by combining the primary keys from two tables which have relations. Normally used in an associative entity'.
- **Purpose** – server as a primary key and used to link (connect) to other tables (ie foreign keys).
- **Example** : Student Number and Course ID



3. Compound Key



Student No.	Course Code	Date		
WM12D07	AACS1304	22-1-24		
WM78B88	AACS3074	04-1-24		
WM64D11	AACS3311	18-1-24		
WM78B88	AACS1304	04-1-24		
WM64D11	AACS3074	18-1-24		
WM12D07	AACS1602	22-1-24		
WM64D11	AACS1304	18-1-24		

Compound Key (serves as ONE primary key for this table)

I wish to request for a replacement of the following :

Course Code	Programme/Tutorial Gp.	Class Type	Population	From				To				
				Date	Day	Time	Venue	Date	Day	Time	Venue	
AMIS1013	DFT1S1G16	tutorial	29	21-08-2024	Wednesday	15:00-16:00	K205	04-09-2024	Wednesday	09:00-10:00	-	<a>Edit <a>Delete
AMIS1013	DFT1S1G16	practical	29	22-08-2024	Thursday	13:00-14:00	K301	04-09-2024	Wednesday	10:00-11:00	-	<a>Edit <a>Delete

Course Code:

Class Type:

From: Date:

Time: To

Venue:

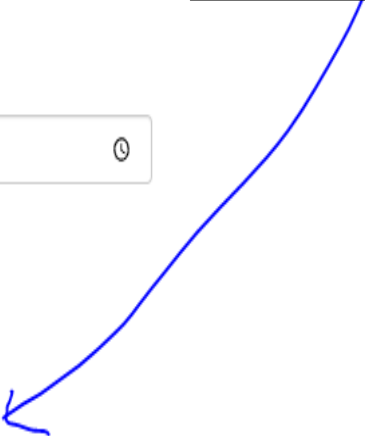
To: Date:

Time: To

Venue:

Programme/Tutorial Group:

Only show one record data



Database Design Language (DBDL)

- One of the design tools used to represent a data table and its data *fields* or *attributes*.

- ERD :

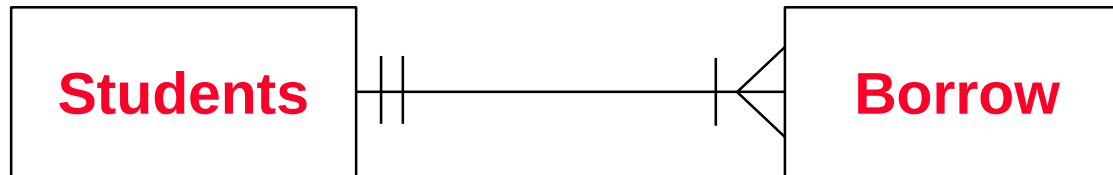


- DBDL : STUDENT (Student ID, Name, Address, Advisor No.*)
- DBDL Components :
 - Entity (Table) Name – in CAPITAL letters.
 - Primary Key – first field or a combination of fields, to be underlined.
 - Non-Key Fields – other data fields describing other non-key fields.
 - Foreign Key (if any) – last field. Indicated with an asterisk (*)

Steps in Database Design

1. Create an **initial** ERD

- Identify 2 entities which are related. This can be carried out using the first 'activity' or 'process'.



2. Assign all data elements to entities

- List **key** and **non-key** data in Database Design Language (**DBDL**) format.



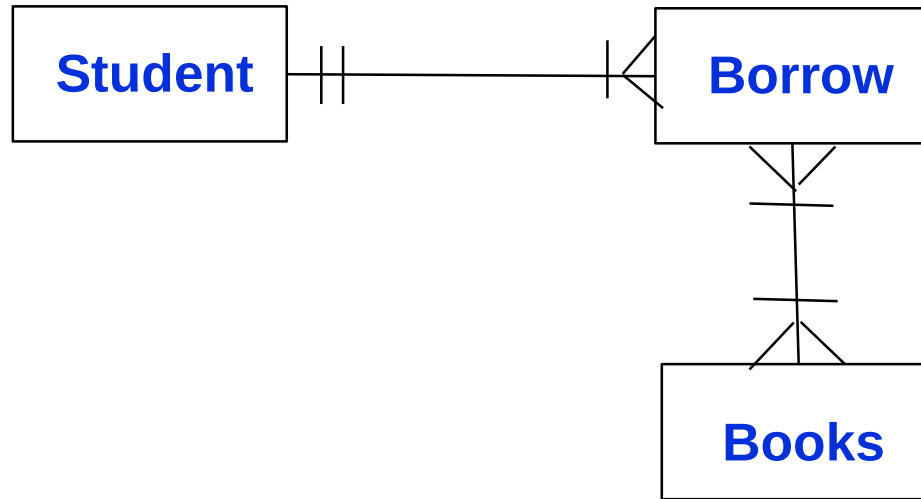
- Example** (in DBDL format)

STUDENT (Student Number, Name, Address, Handphone)

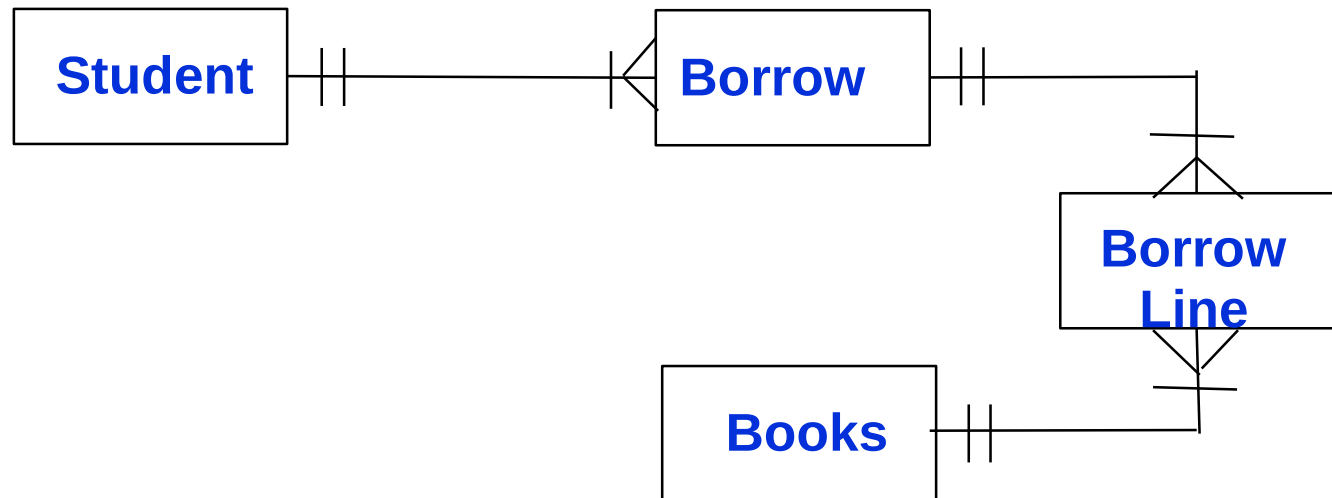
BORROW (Borrow Number, Date, Student ID*)

3. Resolve any many-to-many relationship

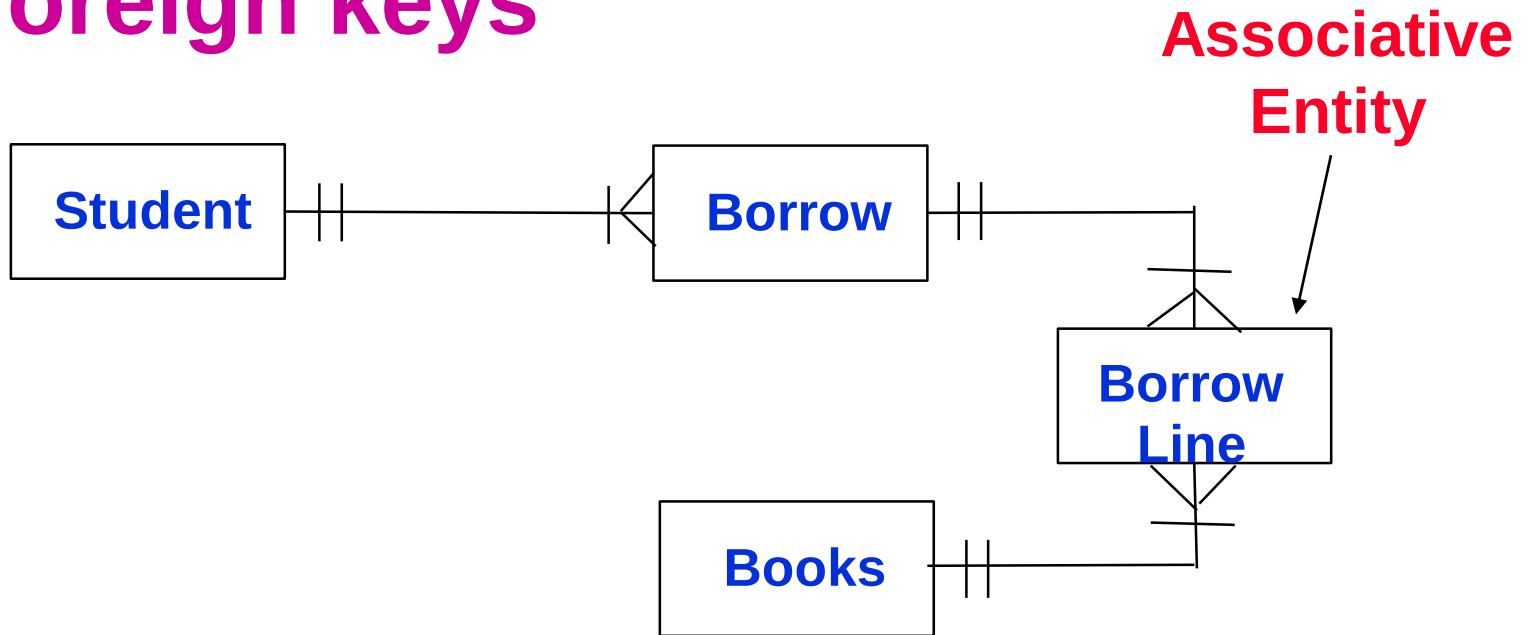
Before



After



4. Relating the Entities by using foreign keys



STUDENT (Student ID, Name, Address, Handphone)

BORROW (Borrow Number, Date, Student ID*)

BOOK (Book Code, Title, Edition, Year-Published)

BORROWLINE (Book Code*, Borrow Number*, Qty)

5. Creating data dictionary entries

- **Data dictionary** – details about **each** data field.
- **Software tools** – eg. Oracle, DB2 can be used.

Data Dictionary

Example of Data Dictionary Format

- **Field name** – the primary name of the data or control item, the data store, or an external entity.
- **Field length** – the number characters
- **Field type** – numeric, alphabetic or alphanumeric
- **Alias** – other names used for the first entry.
- **Field content description** – a notation for representation content.
- **Supplementary information** – other information about: preset values (if known), nullability, restriction or limitation, etc.

Example of Data Dictionary Tools

The screenshot displays the Microsoft Access application window titled "Microsoft Access - [Client : Table]". The main window shows a table with the following fields:

Field Name	Data Type	Description
Client Number	Text	Client Number (Primary Key)
Name	Text	Client Name
Address	Text	Street Address
City	Text	City
State	Text	State (Two-Character Version)
Zip Code	Text	Zip Code (Five-Character Version)
Telephone Number	Text	Telephone number (8 digits)
Client Type	Text	Client type: EDU - Education, MAN - Manufacturing, SER - Services
Amount Paid	Currency	Amount Paid by Client This Year
Current Due	Currency	Current Due from Client This Period
Trainer Number	Text	Number of Client's Trainer

Below the table, the "Field Properties" task pane is open, showing the "General" tab for the selected field. The properties are:

Property	Value
Field Size	4
Format	>
Input Mask	
Caption	
Default Value	
Validation Rule	
Validation Text	
Required	No
Allow Zero Length	Yes
Indexed	Yes (No Duplicates)
Unicode Compression	Yes
IMC Mode	On
IMC Sentence Mode	None
Smart Tag	

On the left side of the image, eight numbered yellow boxes (1-8) have arrows pointing to specific elements in the screenshot:

- 1 points to the "Field Name" column header.
- 2 points to the "Data Type" column header.
- 3 points to the "Description" column header.
- 4 points to the "Field Properties" task pane title.
- 5 points to the "General" tab in the Field Properties task pane.
- 6 points to the "Field Size" property.
- 7 points to the "Validation Rule" property.
- 8 points to the "Required" property.

A text box on the right side of the Field Properties task pane contains the following text:

A field name can be up to 64 characters long, including spaces. Press F1 for help on field names.

At the bottom of the window, the status bar reads: "Design view. F6 = Switch panes. F1 = Help."

Home Relationships Customers		
	Field Name	Data Type
🔑	ID	AutoNumber
	Company	Short Text
	Last Name	Short Text
	First Name	Short Text
	E-mail Address	Short Text
	Job Title	Short Text
	Business Phone	Short Text
	Home Phone	Short Text
	Mobile Phone	Short Text
	Fax Number	Short Text
	Address	Long Text
	City	Short Text
	State/Province	Short Text
	ZIP/Postal Code	Short Text
	Country/Region	Short Text
	Web Page	Hyperlink
	Notes	Long Text
	Attachments	Attachment

Home Relationships Customers		
General Lookup		
Field Size	50	
Format		
Input Mask		
Caption		
Default Value		
Validation Rule		
Validation Text		
Required	No	
Allow Zero Length	No	
Indexed	Yes (Duplicates OK)	
Unicode Compression	Yes	
IME Mode	No Control	
IME Sentence Mode	Phrase Predict	
Text Align	General	

Home

Relationships

Customers

	ID	Company	Last Name	First Name	E-mail Address	Job Title	Business Phone	Home Phone	Mobile Phone	Fax Number	Address
+	1	Company A	Bedecs	Anna		Owner	(123)555-0100			(123)555-0101	123 1st Street
+	2	Company B	Gratacos Solso	Antonio		Owner	(123)555-0100			(123)555-0101	123 2nd Street
+	3	Company C	Axen	Thomas		Purchasing Representative	(123)555-0100			(123)555-0101	123 3rd Street
+	4	Company D	Lee	Christina		Purchasing Manager	(123)555-0100			(123)555-0101	123 4th Street

Uses of Data Dictionary

- **During analysis**
 - helps the analyst **organise** information about the data
 - excellent tool for **communicating** with the user
 - serves as a **memory aid**. Any missing information will alert the analyst
- **During design** - as a source of data about data. Used to generate record, file, and data base formats.
- **During implementation** - serves as a common base for all the programmers working on a system.
- **During maintenance** - gives the maintenance programmer a clear idea of how a particular data element is used.

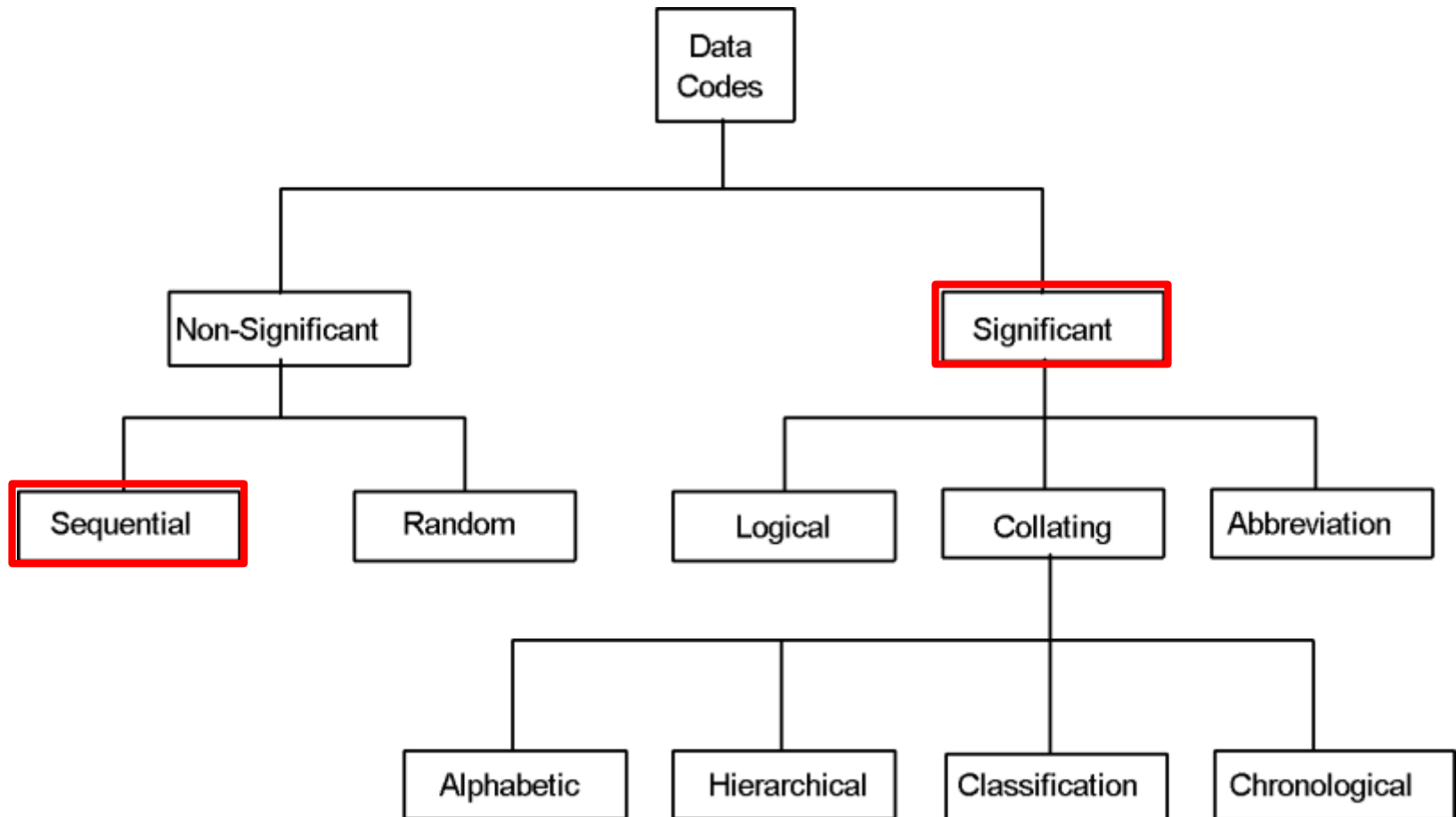
Code Design (Primary Key)

Code Design (Primary Key)

- **Meaning** - code is a set of numbers and letters (called a 'primary key') representing an item (a record)
- **Purpose** - to facilitate **unique** identification and retrieval of items of information.

Types of Codes

Example of code : 54-501306-3487



Types of Codes

- **Sequence Codes**

- These are numbers or letters assigned in a specific order. Sequence codes contain no additional information other than an indication of order of entry into the system.
- **Example** : A HRIS issues consecutive employee numbers to identify employees eg 2056 means number 2056th employee.

- **Block Sequence Codes**

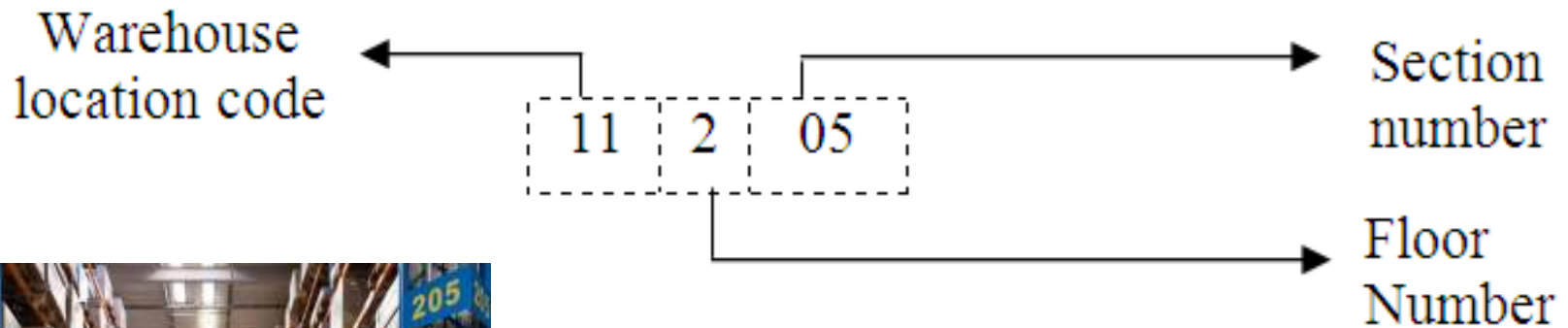
- These codes use blocks of numbers for different classifications. 100-199 , 200-299
- **Example** : College course number. 100-level courses eg. Chemistry 110 and Mathematics 125, are Year 1 courses, whereas 200s indicate Year 2 courses.

Types of Codes

- Significant **Alphabetic Codes**
 - They use alphabet letters to distinguish one item from another based on a category, an abbreviation, or an easy-to-remember value, called a mnemonic code.
 - **Example :**
 - **SW** for software
 - **HW** for hardware
 - **PL** for plastics

Types of Codes

- Significant **Digit** Codes
 - These codes distinguish items by using a series of subgroups of digits.
 - **Example** : inventory location codes



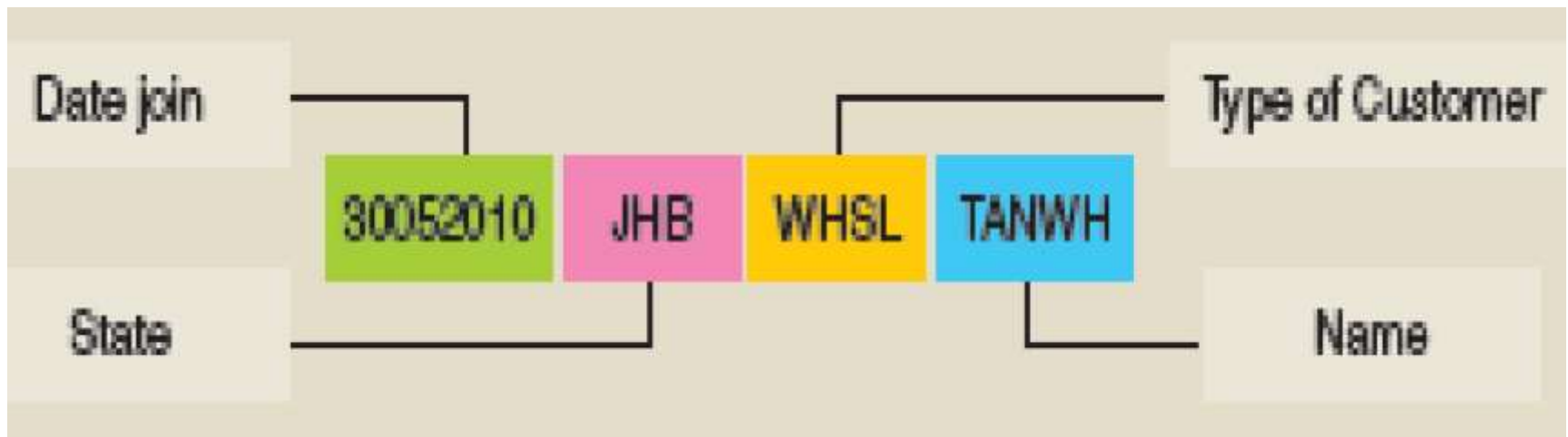
How many Floors can the Code support ?

How many warehouses can the Code support ?

Types of Codes

- **Derivation Codes**

- These codes combine data from item attributes, or characteristics, to build the code.
- **Example** of Customer Number:
(30052010JHBWHSLTANWH)
- 2024KL000000001-2024KL999999999



Students to comment on the major potential problems of using the above Code design

***Additional Info:**

Universal Product Code (UPC)

- **It is a barcode which is uniquely assigned to an inventory item.**
- **The code is used to represent a product in the POS system.**



*Additional Info:

Quick Response (QR) Code

- It is the trademark for a type of matrix barcode (or two-dimensional code) designed for various industries.
- Recently, the system has become popular due to its fast readability and large storage capacity.



Designing a Code (**Primary Key**)

Generally speaking :

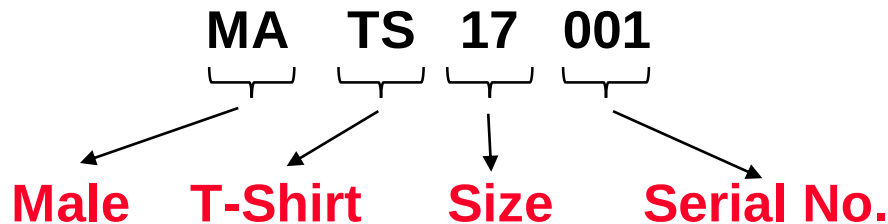
- For **Transaction File** Primary Key – must use *Sequence Code* eg. PO Number, Quotation Number, Invoice Number
- For *complex* **Master File** Primary Key – must use *Significant Code* eg. Stock Code
- For *simple* **Master File** Primary Key – can still use *Sequence Code* eg. Employee No., Customer No.

Exercise for Students

- Suggest and provide example of codes suitable for the following :
 - Customer Number
 - Purchase Order Number
 - Stock Code for clothes

Suggested Answers

- Suggest and provide example of codes suitable for the following :
 - **Customer Number**
 - Use Sequence Code
 - 000001 (allow expandability)
 - **Purchase Order Number**
 - Use Sequence Code
 - 0000000001 (allow expandability)
 - **Stock Code for clothes**
 - Use Significant Code



Characteristics of A Good Code

1. **Uniqueness.** The code has only **one** value for a record in the entity. No other records have the same value. The code is able to identify a record uniquely.
2. **Expandability.** The code should be long enough (**sufficient** characters) to allow for possible future growth eg. use more characters, add serial numbers (end), year or alphabets to the code (front).

2024KL000000001-2024KL999999999

2024SK000000001-2024SK999999999

3. **Conciseness.** The code should be as **short** as possible and still able to represent a record uniquely without sacrificing expandability characteristic.

Characteristics of A Good Code (cont..)

4. **Simplicity.** The code must be simple to remember, understood, interpret and **apply** by the users, particularly workers with little experience eg. use alphabets, M = Medium, L = Large
5. **Stability.** Codes which do not require to be frequently changed or **updated** when record characteristics change. 8digit—9digit
6. **Flexibility.** The code should be easy to **modify** (eg add new characteristic) to reflect necessary changes in characteristics of a record in the entity. 2024KD000000001



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<https://dimerco.com/get-to-know-malaysias-ports/>



Exercise for Students

Consider the following relations and express them in a proper **DBDL** format for a **partially** extracted Shipping Database:

• ~~DESTINATION~~ (Dest Code, Destination)

• ~~SHIP~~ (Ship Id, Model, Weight, Capacity)

• ~~SCHEDULE~~ (Dest Code, Ship Id)

—

Port (PortID, Name, Location, Country)

Ship (ShipID , Name, Type, Capacity, Model, Weight, HomePort ,
RegistrationNumber,)

Schedule(ScheduleID, ShipID , DepartureDate , DepatureTime,
ArrivalDate , ArrivalTime, Status, DeparturePortID, DestinationPortID)

a) Express the above again using a proper DBDL format. Indicate all primary and foreign keys.

b) Draw an ERD to show the above.



Suggested Answers

a) DBDL

Port (PortID, Name, Location, Country)

Ship (ShipID, Name, Type, Capacity, Model, Weight, HomePort* ,
RegistrationNumber,)

Schedule(ScheduleID, ShipID* ,DepartureDate , DepatureTime, ArrivalDate,
ArrivalTime, Status, DeparturePortID*, DestinationPortID*)

b) ERD

- Relationship: One-to-Many (One Ship to Many Schedules)
- Relationship: Many-to-One (Many Schedules to One Departureport)
- Relationship: Many-to-One (Many Schedules to One DestinationPort)

